Md Anwarul Islam Aion

Avionics Engineering, Aviation and Aerospace University, Bangladesh (AAUB)
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Research Interests

Expertise in **radar signal processing**, from building a Pulse-Doppler radar end to end. As well as a strong interest in **wireless communication** and **antenna design**. **Integrated sensing and communication (ISAC)** is where these two combine and that is the motivation. Now working on how a 5G base station's own transmitted signal can be used to sense from its downlink. This is a radar problem inside a communication system. The open question is how much sensing quality survives the scheduler. Also working on **reconfigurable intelligent surfaces** and planning to bring the two together for future wireless communication. Also worked on **5G C-band interference detection and mitigation for FMCW Radar Altimeters** on SDR, factor-graph SLAM for GNSS-denied navigation with computer vision and microwave brain imaging with Vivaldi antennas.

Research Experience

Research Assistant 2026–Present
AAUB Radar Project *(Aviation and Aerospace University, Bangladesh)*
Supervisor: Air Cdre Dr. Mohammad Akhter Jamil, Dean Hours per week: 20

- Appointed by special exception as the **first undergraduate in the university's history** to hold a paid research assistantship.
- Building a Pulse-Doppler radar end to end. The work covers I/Q processing, MTI and **CFAR** detection, real-time acquisition over UDP and integration with a legacy radar display. The target is an FPGA-compatible build.

Undergraduate Thesis — RIS-assisted Integrated Sensing and Communication 2026–2027
Supervisor: Air Cdre Dr. Mohammad Akhter Jamil, Dean

Education

BSc in Avionics Engineering 2023–2027 (Expected July 2027)
Aviation and Aerospace University, Bangladesh (AAUB) *Lalmonirhat, Bangladesh*

- CGPA **3.62/4.00** (through 6th semester), under the supervision of the **Bangladesh Air Force**.
- *Relevant Coursework:* Communication Systems, Digital Signal Processing, Signals & Systems, Fourier & Laplace Transform, Electromagnetic Field Theory, Microwave & Antennas, RADAR Engineering.

Higher Secondary (HSC) — Notre Dame College, Dhaka; **Perfect GPA** 2022
Secondary (SSC) — Kushtia Zilla School; **Perfect GPA**, National Merit Scholar 2020

Research Publications

[1] **Range-Resolution Limits of RIS-Assisted Radar: An Element-Bandwidth Criterion and Its Full-Wave Test**. *IEEE International Conference on Communications (ICC)*, SAC: Integrated Sensing and Communication, 2027. **Submitted**.

[2] **The Cross-Track Information Cost of Weakly Observable Landmark Orientation: A Closed-Form Retention Law and Its Operating Conditions in Visual Teach-and-Repeat**.

Journal of Field Robotics (Wiley), 2026. **Under review**.

[3] **A Framework for Resilient Power Systems in CanSat-class Reusable Small Satellites**. *26th International Middle East Power Systems Conference (MEPCON)*, Egypt, 2025. DOI: 10.1109/MEPCON66918.2026.11360120.

[4] **LIFE-MAP: An Indigenous Biosignature Detection Protocol Using Multi-Biomolecule Based Analysis**. *IEEE Region 10 Conference (TENCON)*, Malaysia, 2025. DOI: 10.1109/TENCON66050.2025.11375044.

Technical Skills

Wireless & 5G	**OpenAirInterface** (gNB, core), **5G NR** (TDD, SSB, CSI-RS, PDSCH, scheduling), **OFDM radar / ISAC**, MATLAB 5G Toolbox
Radar & Signal Processing	Matched filtering, **range–Doppler** processing, **MTI** and **CFAR** detection, I/Q acquisition chains, **MATLAB**/Simulink, **GTSAM**
RF & Electromagnetics	**CST Studio Suite** (full-wave unit-cell and 128-element panel), RIS and antenna design, LTspice
SDR & Embedded	**ADALM-Pluto** (libiio), **GNU Radio**, USRP B210, real-time **C++** (CMake, ring buffers, UDP), **FreeRTOS**, Altium Designer
Programming & Tools	**Python** (NumPy, SciPy, scikit-learn, OpenCV), **C++**, **LaTeX**, pytest

Engineering Projects

[1] **Team Lead (2026–Present); Electrical Engineer** 2024–2025
AAUB CanSat *[Team Ababil]* *Lalmonirhat, Bangladesh*
- Electrical and power-systems engineer for a CanSat-class satellite with RF telemetry. The team became the country's first **Global Finalist** at CanSat 2025 in **Virginia, USA** (AAS & NASA).

[2] **Science Module Lead** Nov 2024–Apr 2026
AAUB Mars Rover *[Team Rover71]* *Lalmonirhat, Bangladesh*
- Built the Science & Environment Analysis subsystem from scratch: sensing payload, instrument chain, data handling. The team placed **8th globally** in the Anatolian Rover Challenge 2025.

[3] **Team Lead** 2024–2025
Oblique-Wing UAV Project *[Independent]* *Bangladesh*
- Designed and flew an oblique-wing prototype after the **NASA AD-1** configuration. It used a custom 3D-printed EDF propulsion unit.

Grants, Awards & Honours

Research Grants — **AAUB**: Mars rover; habitability analysis module 2023–2025
Global Finalist — CanSat Competition 2025, Virginia, USA (AAS & NASA) 2025
Global Finalist — Anatolian Rover Challenge 2025, Turkey (8th globally) 2025
National Champion — Four national debate championships, Bangladesh 2019–2022

References

Name	Employer	Title	Email
Air Cdre Dr. Mohammad Akhter Jamil (Thesis Supervisor)	AAUB	Dean	jamil8703@gmail.com
Air Cdre (Retd) Dr. Md Afzal Hossain (Project Supervisor)	AAUB	Distinguished Professor	afzal@aaub.edu.bd
Dr. Md-Mizanur Rahoman (Research Supervisor)	Begum Rokeya University	Professor	mizan@brur.ac.bd